

Bayesian Linear Travel-Time Tomography: A Controlled Numerical Study of Reconstruction and Uncertainty Calibration

Abstract

We study a finite-dimensional Bayesian inverse problem motivated by crosshole seismic travel-time tomography. The unknown subsurface slowness field is discretized on a uniform square grid inside a square domain, while travel-time measurements are modeled as line integrals along known straight rays. This gives a linear forward model

$$\mathbf{d} = A\mathbf{s} + \boldsymbol{\varepsilon}.$$

We place a Gaussian prior on the discretized slowness field using a squared-exponential covariance, and assume Gaussian observation noise. The resulting posterior distribution is available in closed form.

The project has two goals. First, we construct and validate the complete forward and Bayesian inversion pipeline. Second, we investigate uncertainty calibration in a controlled synthetic setting. In particular, we distinguish between a correctly specified Bayesian generative model, where the true field is drawn from the assumed prior, and deliberately misspecified experiments in which the prior, noise level, or true field is inconsistent with the inference model.

The resulting study is intended as a small, reproducible computational project rather than as a claim of a new tomography algorithm. Its main purpose is to demonstrate the relationship between acquisition geometry, regularization, reconstruction accuracy, and posterior uncertainty.

1 Problem statement

We consider a two-dimensional square domain

$$\Omega = [0, W] \times [0, W].$$

The unknown medium is described by its slowness field

$$s(\mathbf{x}) = \frac{1}{v(\mathbf{x})}, \quad \mathbf{x} \in \Omega,$$

where $v(\mathbf{x})$ denotes wave speed.

Under the straight-ray approximation, the travel time associated with a ray $L_i \subset \Omega$ is

$$t_i = \int_{L_i} s(\mathbf{x}) d\ell.$$

We place sources along the left boundary of the domain and receivers along the right boundary. Every source is connected to every receiver by a straight line segment. The resulting collection of rays provides the acquisition geometry.

The straight-ray approximation is adopted deliberately. In a fully physical treatment, rays may depend on the slowness field through Fermat's principle, resulting in a nonlinear forward problem. Here the ray paths are fixed in advance so that the inverse problem remains linear.

2 Discretization

Partition Ω into a uniform $n \times n$ grid of square cells, giving n^2 cells total. The grid is uniform in both directions: cell width equals cell height. This eliminates cell aspect ratio as a variable, so that any anisotropy observed in the reconstruction or in posterior uncertainty is attributable only to the acquisition geometry and the prior.

We approximate the slowness field by a piecewise constant function:

$$s(\mathbf{x}) \approx s_j \quad \text{inside cell } j.$$

For ray i , let A_{ij} denote the length of the intersection of ray L_i with cell j . The discretized travel time is therefore

$$t_i = \sum_{j=1}^n A_{ij} s_j.$$

Writing $\mathbf{s} = (s_1, \dots, s_n)^\top$ and collecting the travel times into a vector $\mathbf{t} = (t_1, \dots, t_m)^\top$, we obtain the linear forward model

$$\mathbf{t} = A\mathbf{s}, \quad A \in \mathbb{R}^{m \times n}.$$

The matrix A depends only on the acquisition geometry and the discretization. It is therefore constructed once before the inverse problem is solved, and should be validated directly: for every ray i , $\sum_j A_{ij}$ must equal the total length of that ray, up to numerical precision (see Section 7).

3 Observation model

The measured travel times are modeled as

$$\mathbf{d} = A\mathbf{s}_{\text{true}} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \Sigma_d).$$

For the initial experiment we use independent, identically distributed Gaussian timing errors:

$$\Sigma_d = \sigma^2 I_m.$$

Thus $\mathbf{d} \mid \mathbf{s} \sim \mathcal{N}(A\mathbf{s}, \Sigma_d)$. The parameter σ controls the magnitude of measurement noise and will be varied in the sensitivity experiments.

4 Gaussian prior

The discretized slowness vector is assigned a Gaussian prior

$$\mathbf{s} \sim \mathcal{N}(\mathbf{s}_0, C_s).$$

The prior mean is chosen to represent a homogeneous background medium, $\mathbf{s}_0 = s_{\text{bg}} \mathbf{1}$.

For the primary experiment we use a squared-exponential covariance. Let $\mathbf{x}_j$ denote the center of cell j . Define

$$(C_s)_{jk} = \tau^2 \exp\left(-\frac{\|\mathbf{x}_j - \mathbf{x}_k\|^2}{2\ell^2}\right) + \epsilon \delta_{jk},$$

where:

- τ^2 controls the marginal variance of the prior;
- ℓ is the spatial correlation length;
- ϵ is a small numerical jitter term added to the diagonal (e.g. $\epsilon \sim 10^{-6}\tau^2$).

Numerical stability note

The squared-exponential covariance is positive definite in exact arithmetic for any set of distinct grid points, but in floating-point practice it is frequently *numerically* near-singular — particularly at fine grid spacing or large ℓ , where many rows of C_s become nearly identical. Without the jitter term ϵ , Cholesky factorization or direct inversion of C_s (or of C_s^{-1} , as required by the posterior update in Section 5) will intermittently fail or produce badly conditioned results. The jitter term is standard practice in Gaussian process computation and should be treated as a required part of the implementation, not an optional safeguard.

The length scale ℓ controls the smoothness assumed by the inversion. Small ℓ permits relatively rapid spatial variation, whereas large ℓ strongly favors smooth fields.

5 Posterior distribution

Because the likelihood and prior are both Gaussian and the forward map is linear, the posterior distribution is Gaussian:

$$\mathbf{s} \mid \mathbf{d} \sim \mathcal{N}(\mathbf{s}_{\text{post}}, C_{\text{post}}).$$

The posterior covariance is

$$C_{\text{post}} = \left(A^\top \Sigma_d^{-1} A + C_s^{-1} \right)^{-1},$$

and the posterior mean is

$$\mathbf{s}_{\text{post}} = C_{\text{post}} \left(A^\top \Sigma_d^{-1} \mathbf{d} + C_s^{-1} \mathbf{s}_0 \right).$$

For the independent-noise model $\Sigma_d = \sigma^2 I$, these become

$$C_{\text{post}} = \left(\frac{1}{\sigma^2} A^\top A + C_s^{-1} \right)^{-1}, \quad \mathbf{s}_{\text{post}} = C_{\text{post}} \left(\frac{1}{\sigma^2} A^\top \mathbf{d} + C_s^{-1} \mathbf{s}_0 \right).$$

Thus the entire inverse problem can be solved without iterative optimization or Monte Carlo sampling. In implementation, both quantities should be computed via a Cholesky factorization of $(A^\top \Sigma_d^{-1} A + C_s^{-1})$ and triangular solves, rather than an explicit matrix inverse, for numerical stability.

6 What is being investigated?

The primary computational question is:

How do acquisition geometry, observation noise, and prior correlation length affect reconstruction accuracy and posterior uncertainty in a linear travel-time tomography problem?

A second question concerns calibration:

When does the posterior covariance provide a quantitatively meaningful description of reconstruction uncertainty?

These questions are investigated in a controlled synthetic setting in which the true slowness field is known.

7 Experiment I: construction and validation

The first stage is simply to construct the complete tomography pipeline.

Step 1: define the domain

Choose $W > 0$ and construct a uniform $n \times n$ grid of square cells. For an initial implementation, $n = 20$ is sufficient.

Step 2: define sources and receivers

Place N_s sources uniformly along the left boundary $x = 0$ and N_r receivers uniformly along the right boundary $x = W$. Every source is connected to every receiver, giving $m = N_s N_r$ rays.

Step 3: construct the sensitivity matrix

For each source–receiver pair, construct the corresponding straight line segment. For every grid cell, compute the length of the ray segment contained in that cell. These intersection lengths form the entries of A : $A_{ij} = \text{length}(L_i \cap \text{cell } j)$. As a basic validation, verify numerically that for every ray, $\sum_j A_{ij}$ equals the total length of that ray, up to numerical precision.

Step 4: define a synthetic truth

For the first reconstruction experiment, choose a smooth deterministic slowness field. One convenient choice is a homogeneous background plus a localized Gaussian anomaly:

$$s_{\text{true}}(x, y) = s_{\text{bg}} + \delta s \exp\left(-\frac{(x - x_0)^2 + (y - y_0)^2}{2r^2}\right).$$

The corresponding cell values define $\mathbf{s}_{\text{true}} \in \mathbb{R}^n$. This deterministic truth is useful for directly measuring reconstruction error, and is used again in Experiment III below.

Step 5: generate synthetic data

Compute noiseless travel times $\mathbf{t}_{\text{true}} = A\mathbf{s}_{\text{true}}$. Then draw $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2 I)$ and form $\mathbf{d} = \mathbf{t}_{\text{true}} + \boldsymbol{\varepsilon}$.

Step 6: compute the posterior

Given $\mathbf{d}, A, \mathbf{s}_0, C_s, \sigma$, compute $\mathbf{s}_{\text{post}}$ and C_{post} . The main visualization should show: (1) the true field $\mathbf{s}_{\text{true}}$; (2) the posterior mean $\mathbf{s}_{\text{post}}$; (3) the posterior standard deviation $\sqrt{\text{diag}(C_{\text{post}})}$.

This establishes that the forward model and inverse calculation are working before any calibration experiment is attempted.

8 Experiment II: Bayesian calibration under the correctly specified model

The second experiment provides a clean statistical benchmark.

Instead of fixing $\mathbf{s}_{\text{true}}$, generate the true field from the same prior used by the inversion: $\mathbf{s}_{\text{true}} \sim \mathcal{N}(\mathbf{s}_0, C_s)$. Conditional on this realization, generate observations according to $\mathbf{d} \mid \mathbf{s}_{\text{true}} \sim$

$\mathcal{N}(A\mathbf{s}_{\text{true}}, \Sigma_d)$. The inference model uses exactly the same C_s and Σ_d . This is the correctly specified Bayesian generative model.

For each independent realization:

1. draw $\mathbf{s}_{\text{true}} \sim \mathcal{N}(\mathbf{s}_0, C_s)$;
2. generate $\mathbf{d} = A\mathbf{s}_{\text{true}} + \boldsymbol{\varepsilon}$;
3. compute $\mathbf{s}_{\text{post}}$ and C_{post} ;
4. compute standardized errors;
5. compute the global Mahalanobis error.

For each cell define

$$z_j = \frac{s_{\text{true},j} - s_{\text{post},j}}{\sqrt{(C_{\text{post}})_{jj}}}.$$

Under the correctly specified Gaussian model, the posterior error conditional on the observed data satisfies $\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}} \mid \mathbf{d} \sim \mathcal{N}(0, C_{\text{post}})$. Consequently,

$$Q = (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}})^\top C_{\text{post}}^{-1} (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}})$$

satisfies $Q \sim \chi_n^2$. This provides an exact calibration benchmark: because the generative model and the inference model agree by construction, this experiment tests software correctness rather than any deeper statistical claim, and is essentially a closed-form instance of simulation-based calibration (cf. Talts et al., 2018). If Q does not follow χ_n^2 here, the bug is in the implementation, not in the statistics.

The numerical experiment should compare the empirical distribution of Q over many independent realizations with the theoretical χ_n^2 distribution.

9 Experiment III: deterministic truth and repeated observation noise

The previous experiment tests calibration under the complete Bayesian generative model, and by construction cannot fail unless the implementation is wrong. A different and practically important experiment is to hold the physical truth fixed and ask whether the posterior is still a trustworthy description of uncertainty — this is now a genuine statistical question, since nothing guarantees the answer is yes.

Choose a deterministic $\mathbf{s}_{\text{true}}$ and generate independent observations

$$\mathbf{d}^{(k)} = A\mathbf{s}_{\text{true}} + \boldsymbol{\varepsilon}^{(k)}, \quad \boldsymbol{\varepsilon}^{(k)} \sim \mathcal{N}(0, \sigma^2 I).$$

For each realization k , compute the corresponding posterior. This experiment answers a different question:

How well does the Bayesian posterior describe uncertainty around a particular fixed physical model when that model is not itself regarded as a random draw from the prior?

The same statistics can be collected:

$$z_j^{(k)} = \frac{s_{\text{true},j} - s_{\text{post},j}^{(k)}}{\sqrt{(C_{\text{post}}^{(k)})_{jj}}}, \quad Q^{(k)} = (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}}^{(k)})^\top (C_{\text{post}}^{(k)})^{-1} (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}}^{(k)}).$$

Importantly, there is no general reason for $Q^{(k)}$ to follow χ_n^2 in this fixed-truth experiment. The result depends on how the deterministic truth relates to the assumed prior.

Two fixed truths, not one

This experiment should be run **twice**, with two qualitatively different deterministic truths, so that the comparison is informative rather than a foregone conclusion:

1. **A smooth truth**, e.g. the Gaussian-anomaly field from Experiment I (Step 4). This field is close to a typical draw from the assumed prior, so the posterior is expected to remain well calibrated (empirical coverage close to nominal, $Q^{(k)}$ close to χ_n^2).
2. **A sharp or discontinuous truth**, e.g. a checkerboard pattern or a rectangular block anomaly with hard edges. This field is atypical under a smooth squared-exponential prior, so the posterior is expected to *under-estimate* uncertainty near the discontinuities, giving poor empirical coverage even though the posterior is mathematically well defined.

Running both cases side by side demonstrates the point of the experiment directly: a posterior can be exactly correct in the generative sense (Experiment II) while being a poor description of uncertainty for a specific physical truth the prior does not represent well (Experiment III, sharp case). This distinction — Bayesian calibration under the generative model versus performance on a fixed physical truth — is the central methodological contribution of the project.

10 Experiment IV: controlled prior misspecification

We next deliberately use different correlation lengths for generating the truth and performing inference. Let C_s^{true} be constructed using a correlation length ℓ_{true} , while the inversion uses C_s^{infer} constructed with $\ell_{\text{infer}} \neq \ell_{\text{true}}$. The data are generated according to $\mathbf{s}_{\text{true}} \sim \mathcal{N}(\mathbf{s}_0, C_s^{\text{true}})$ and $\mathbf{d} = \mathbf{A}\mathbf{s}_{\text{true}} + \boldsymbol{\varepsilon}$, but the posterior is computed using C_s^{infer} .

We then investigate how calibration changes as $\ell_{\text{infer}}/\ell_{\text{true}}$ moves away from 1. This experiment illustrates that a mathematically well-defined posterior need not be a perfectly calibrated representation of physical uncertainty when the prior is misspecified.

11 Experiment V: noise-level misspecification

A second controlled mismatch concerns the observation noise. Generate observations with $\boldsymbol{\varepsilon}_{\text{true}} \sim \mathcal{N}(0, \sigma_{\text{true}}^2 I)$, but perform inference using $\Sigma_{d,\text{infer}} = \sigma_{\text{infer}}^2 I$. We vary the ratio $\sigma_{\text{infer}}/\sigma_{\text{true}}$ and measure its effect on:

- reconstruction error;
- posterior standard deviations;
- standardized residuals;

- Mahalanobis distances;
- empirical coverage of posterior credible intervals.

This provides a simple demonstration of the effect of incorrectly specified measurement uncertainty.

12 Sensitivity to acquisition geometry

The same framework can be used to vary the number of rays. For example, repeat the experiment for several choices of (N_s, N_r) . Increasing the number of source–receiver pairs changes the sensitivity matrix A , which changes the information supplied by the likelihood.

For each geometry we record:

- $\text{rank}(A)$ and the singular values of A ;
- the condition number and singular values of the *regularized* system $A^\top \Sigma_d^{-1} A + C_s^{-1}$, not of A alone. A is typically rank-deficient by construction — many cells are crossed by few or no rays — so its raw singular values understate the information actually available to the posterior once the prior’s regularization is included. The conditioning of the regularized system is the quantity that actually governs C_{post} ;
- the resulting posterior uncertainty $\text{diag}(C_{\text{post}})$.

The objective is not merely to obtain visually sharper images, but to investigate how acquisition geometry affects the information content of the inverse problem.

13 Performance metrics

The project will use several complementary metrics.

Reconstruction error

$$E_{\text{rel}} = \frac{\|\mathbf{s}_{\text{post}} - \mathbf{s}_{\text{true}}\|_2}{\|\mathbf{s}_{\text{true}}\|_2}.$$

Posterior uncertainty

For each cell, $\sigma_{\text{post},j} = \sqrt{(C_{\text{post}})_{jj}}$. These values provide a spatial map of posterior uncertainty.

Standardized error

$$z_j = \frac{s_{\text{true},j} - s_{\text{post},j}}{\sigma_{\text{post},j}}.$$

For a correctly specified model, these quantities should exhibit the behavior predicted by the Gaussian posterior.

Mahalanobis error

$$Q = (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}})^\top C_{\text{post}}^{-1} (\mathbf{s}_{\text{true}} - \mathbf{s}_{\text{post}}).$$

Under the correctly specified Bayesian generative experiment, $Q \sim \chi_n^2$.

Credible-region coverage

For a nominal level $1 - \alpha$, construct marginal credible intervals and measure their empirical coverage over repeated synthetic experiments. The marginal interval for cell j is

$$s_{\text{post},j} \pm z_{1-\alpha/2} \sqrt{(C_{\text{post}})_{jj}},$$

where $z_{1-\alpha/2}$ is the corresponding standard-normal quantile. The empirical fraction of experiments for which $s_{\text{true},j}$ lies inside this interval can be compared with the nominal coverage level.

14 Numerical implementation

The implementation should be divided into independent components:

1. **Grid construction:** create the domain and cell centers.
2. **Ray construction:** generate source–receiver pairs and the corresponding straight line segments.
3. **Ray–cell intersection:** compute A_{ij} for every ray and cell.
4. **Prior construction:** construct the squared-exponential covariance C_s , including the numerical jitter term.
5. **Synthetic data generation:** generate $\mathbf{s}_{\text{true}}$ (both smooth and sharp variants), noiseless travel times, and noisy observations.
6. **Bayesian inversion:** compute $\mathbf{s}_{\text{post}}$ and C_{post} via Cholesky factorization and triangular solves.
7. **Diagnostics:** calculate reconstruction errors, posterior uncertainties, standardized errors, and Mahalanobis distances.
8. **Visualization:** produce maps of the true field, posterior mean, posterior standard deviation, and reconstruction error.
9. **Repeated experiments:** repeat the stochastic experiment over many independent noise/truth realizations.
10. **Sensitivity analysis:** vary ray density, noise level, and prior correlation length.

15 Minimal experimental schedule

To keep the project computationally small and reproducible, the initial study will use the following sequence:

1. Implement a 20×20 grid.
2. Use a modest set of uniformly distributed sources and receivers.
3. Construct and visualize A .
4. Verify ray lengths using row sums of A .

5. Define a smooth deterministic anomaly.
6. Generate one noisy data set.
7. Compute the posterior.
8. Produce the three primary images: true field, posterior mean, posterior standard deviation.
9. Repeat the correctly specified Bayesian experiment (Experiment II) for approximately 100 independent realizations.
10. Plot the empirical distribution of the Mahalanobis statistic against the χ_n^2 reference distribution.
11. Repeat with a deterministic fixed truth (Experiment III), using *both* the smooth anomaly and a sharp/checkerboard truth.
12. Perform a small prior-length-scale sensitivity sweep (Experiment IV).
13. If time permits, add noise misspecification (Experiment V).

16 Expected outputs

The project should produce the following concrete outputs:

1. A reproducible construction of the tomography sensitivity matrix A , validated against known ray lengths.
2. A numerical implementation of the exact Gaussian posterior.
3. A reconstruction of a synthetic slowness field from noisy travel-time measurements.
4. Spatial maps of posterior uncertainty.
5. Empirical calibration results under the correctly specified Bayesian model.
6. A comparison between the correctly specified Bayesian experiment and the fixed-truth experiment, for both a smooth and a sharp fixed truth.
7. At least one controlled misspecification experiment (prior or noise).
8. A short discussion of the relationship between acquisition geometry, prior assumptions, reconstruction quality, and uncertainty calibration.

17 Scope and interpretation

This project is intentionally a controlled linear model rather than a full seismic inversion. The following effects are excluded from the initial study: curved rays; nonlinear dependence of ray paths on the slowness field; attenuation; anisotropy; frequency-dependent effects; wave-equation simulation; non-Gaussian likelihoods.

The purpose of these simplifications is to isolate the relationship between linear tomography, Bayesian regularization, and uncertainty quantification. The resulting model is sufficiently simple that the posterior can be computed exactly. This makes it possible to distinguish errors caused by numerical implementation from errors caused by statistical misspecification.

18 Final objective

The final objective is not simply to demonstrate that Bayesian linear tomography can reconstruct a synthetic image. Rather, the project should demonstrate the complete chain

geometry $\rightarrow A \rightarrow$ synthetic observations $\rightarrow$ Bayesian inversion $\rightarrow$ uncertainty $\rightarrow$ calibration

in a controlled setting. The central comparison is between a *correctly specified Bayesian model* and a *fixed or deliberately misspecified physical truth*. This provides a compact computational investigation of when a posterior distribution is not merely mathematically well defined, but also a useful quantitative description of uncertainty in a tomography problem.